



Chips Off The New Block For Changing Times

In India, developing microprocessors and microcontrollers indigenously is essential to reduce the dependence on imports and make the nation an electronics design and manufacturing hub. As technology continues to evolve, designers need to stay updated to reduce cost, board space and design time for their projects

Ayushee Sharma

With advancements in technology, the industry is quickly moving to new trends every day. Design complexity increases as designers have to ensure that their products have the necessary features to be suitable for numerous applications and stay relevant for a long time in the rapidly-changing market. Without the right design, changes have to be made in the hardware later on, which results in a huge waste of time and cost.

Microprocessor, being the central processing unit, is the most essential design component for most electronic devices. Designers have to often make trade-offs between various factors for reliable and optimised performance. Complexity rises with increasing density of transistors. Through high-level synthesis method, many designers use programming to figure out the hardware required for a specific application.

Most of the pins on a processor are for power supply and ground. If voltage is too high, subsequent overheating can take place.

The clock is also an important consideration to keep all functions in sync and maintain energy efficiency. Although high speed is necessary, processors usually run on a frequency of 3GHz to 4.5GHz so that even the slowest component can complete its operation accurately within the required clock cycle.

Impact of AI

As the focus is shifting to artificial intelligence (AI) processing for features like voice assistants and image processing, the processor needs to be faster, flexible and suited for real-time applications. This requires highly integrated chip manufacturing for meeting computational requirements.

Thermal stress can cause the chip to break if power is not uniform across the chip. System software for such processors must be capable of extracting the neural network from machine learning (ML) frameworks like PyTorch and TensorFlow.

